AMAS: Adaptive Multi-Agent Search for Multi-Hop QA with a Calibrated Conformal Routing Gate

System Report

Abstract

AMAS is a multi-hop question answering system that routes each query between a single-shot answer (SAS) lane and a multi-agent search (MAS) lane. The SAS lane is a single GPT-4o-mini probe over top-$k = 5$ retrieved passages. The MAS lane is an LLM-orchestrated topology over specialized agents (Qwen3-14B for the orchestrator, GPT-4o-mini for subagents). Routing is governed by a foreign verifier used as a split-conformal gate. On HotpotQA, 2WikiMultihop, and Bamboogle, the conformal gate gives a useful cost–quality tradeoff: it keeps 40.5–66.4% of questions in the cheap SAS lane and reduces token cost by 30–55% relative to a matched untrained HERA baseline. The MuSiQue result is the main failure case. AMAS-conformal reaches only 0.104 EM, 0.193 F1, and 0.133 Acc/contain on MuSiQue; even the ungated MAS lane reaches only 0.112 EM and 0.209 F1. The system therefore shows real *compute adaptivity* (routing changes cost), but only partial *quality adaptivity*: the gate often sends easier questions to SAS, yet the underlying MAS lane is too weak on MuSiQue for the adaptive policy to recover strong final accuracy.

1 Introduction

Multi-agent retrieval-augmented generation (RAG) systems decompose a question into agent-specific subroutines—query decomposition, query rewriting, retrieval, evidence selection, context validation, answer generation, reflection, and conclusion—and orchestrate them via an LLM-based controller. HERA [1] is the canonical example: a learned orchestrator selects a topology over specialized agents and runs them sequentially. HERA typically calls between five and eight agents per question, irrespective of question difficulty.

A simpler alternative is the single-agent scaffold (SAS) of Tran and Kiela: one retrieve-then-answer call per question. SAS is cheap (∼1k tokens per question) but loses substantial quality on multi-hop benchmarks compared to a fully orchestrated multi-agent system.

AMAS occupies the middle ground. It runs a one-shot probe before any multi-agent orchestration. A separately calibrated verifier decides whether the probe answer is sufficient. If sufficient, AMAS commits the probe answer (SAS lane) and skips the entire multi-agent stack. If not, AMAS proceeds to run a single MAS turn with a topology sampled from an experience library (MAS lane). The verifier is calibrated by split-conformal prediction [2], which gives a marginal coverage guarantee $P[\text{commit} \wedge \text{wrong}] \leq \alpha$ over exchangeable test items.

This report describes the system as it stands at commit `e54070389`[1] and presents the headline 1000-question evaluation.

[1]The `ledger.py` module exists in the repository but is not wired into agent execution; we therefore omit any discussion of cross-turn memory in this report and treat it as future work.

2 System architecture

High-level pipeline. For each question q, AMAS executes the following pipeline:

1. **Turn 0 – probe.** Retrieve top-$k = 5$ passages $P_0 = \{p_1, \ldots, p_k\}$. Issue a single GPT-4o-mini call with (q, P_0) and receive a candidate answer span $\hat{a}$.
2. **Turn-0 gate.** Run a verifier on $(q, \hat{a}, P_0)$ to produce a real-valued score s. Compare s against a calibrated threshold τ_{high}:
   - if $s \geq \tau_{\text{high}}$: commit $\hat{a}$ as the final answer (SAS lane). Pipeline ends.
   - if $s \leq -\tau_{\text{high}}$: emit a strong refusal and trigger topology mutation in MAS.
   - otherwise: continue to MAS.
3. **Turn 1 – MAS.** The orchestrator emits a topology over the eight available agents (selected agents, execution order, dependencies). Agents are executed in topological order with parallel same-step batches. The `ConcludeAgent` (or its fallback) emits the final answer span.
4. **Score and return.** Compute EM, token-F1, contain, and total token count.

The pipeline runs $T_{\text{max}} = 1$ MAS turn after the probe, matching the depth used by HERA in the published baseline. Adaptivity in this configuration is purely lateral: the gate decides between SAS and MAS, but the depth of MAS is fixed.

Eight agents. The agent inventory and roles are summarized in Table 1. Each LLM agent has a fixed role description, an instruction block, a list of operational rules, and a list of behavioral principles. Agents emit JSON outputs validated by a lenient parser. The `Retriever` is a tool agent (no LLM call) that wraps an E5-Flat dense index over wiki18.

Agent	**Role**
`QueryDecomposer`	Decompose a multi-hop question into 1–4 atomic single-hop sub-questions.
`QueryRewriter`	Rewrite or expand the question into one or more search-friendly queries.
`Retriever`	Tool agent: retrieve top-k passages from wiki18 for each query.
`EvidenceSelector`	Select the minimal set of passages that are sufficient to answer the question.
`ContextValidator`	Verify whether the assembled context is sufficient to answer the question.
`AnswerGenerator`	Produce a concise final answer span from the question and selected passages.
`ReflectAgent`	Critique a proposed answer for errors, missing reasoning, or contradictions.
`ConcludeAgent`	Aggregate upstream agent outputs and emit the final answer span.

Table 1: The eight specialized agents of the AMAS MAS lane. All agents except the `Retriever` are GPT-4o-mini calls.

Orchestrator and experience library. The orchestrator is a Qwen3-14B model that emits a JSON topology object given the question, an inferred query profile (e.g., `bridge`, `comparison`, `temporal`, `intersection`), and the top-k retrieved *insights* from an experience library. Each insight is a short natural-language rule keyed by query profile. The library $\mathcal{E} = \{(c_i, z_i, u_i)\}_{i=1}^{N}$ stores at most 30 insights at a time and is updated by the paper-faithful Algorithm 3 (`ADD/MERGE/PRUNE/KEEP`).

The orchestrator output is normalized by `validate_topology()`, which rejects topologies with more than eight steps, missing dependencies, or invalid agents and falls back to a default topology when validation fails.

Probe. The probe is a single GPT-4o-mini call with the prompt template:

> *You are a single-shot QA agent. Read the question and the retrieved passages, then output a SHORT ANSWER SPAN. One entity, one date, one short phrase, or yes/no. NEVER write a sentence. NEVER explain. Cap at 8 words. Bare span only — no quotes, no period, no preamble.*

The probe returns a JSON object `{"answer": str, "confidence": float}`. At inference we use a probe group size $G = 1$, i.e., one probe call per question, and pass the consensus answer to the gate. The probe shares the same retrieval pool as the rest of the pipeline.

Conformal verifier (Route A). The verifier is a separate GPT-4o-mini call with no chain-of-thought from the answering agent. Given $(q, \hat{a}, \text{evidence})$ it emits `{"verdict": "YES|NO", "confidence": float}`. We convert this into a real-valued score

$$s = \begin{cases} \log \frac{c}{1-c} & \text{verdict} = \text{YES} \\ -\log \frac{c}{1-c} & \text{verdict} = \text{NO} \end{cases} \tag{1}$$

where c is the confidence (clipped to $(\varepsilon, 1 - \varepsilon)$). This is a logit-style transform of the verifier's confidence.

The threshold τ_{high} is chosen by split-conformal prediction on a held-out calibration set $\mathcal{C}$ of size n. Let $s^{\text{cor}}_{(1)} \leq s^{\text{cor}}_{(2)} \leq \dots$ be the sorted verifier scores on the calibration items where the probe answer was *correct* (EM or contain match against the gold). For a target SAS-error rate α, we set

$$\tau_{\text{high}} = s^{\text{cor}}_{(\lceil (1-\alpha)(n+1) \rceil)}. \tag{2}$$

Under exchangeability, this gives the marginal guarantee $P[\text{commit} \wedge \text{wrong}] \leq \alpha$.

For the experiments below, $\alpha = 0.05$ and the calibration set is the 199 verifier-scored questions from the IRCoT training pool (id-disjoint from the test sets), giving $\tau_{\text{high}} = 6.91$.

A smaller threshold $\tau_{\text{low}} =$ 25th percentile of incorrect calibration scores defines a defer band. Within the defer band the gate issues a single re-verification before continuing. A score $s \leq -\tau_{\text{high}}$ is treated as a strong refusal and triggers MAS topology mutation.

Bayesian gate (Route B, ablation). A second gate is included as a closed-system ablation that uses no foreign verifier. We tested two formulations.

Route B v1 (entropy-only). The original closed-system gate commits when the belief-state entropy alone is below a threshold:

$$\text{commit if } H(\text{belief}) \leq \lambda^{-1}. \tag{3}$$

This formulation is degenerate at probe group size $G = 1$: a single probe sample produces a single candidate in the belief, so $H(\text{belief}) = 0$ trivially and the gate commits on every question regardless of λ. Section 4.4.2 confirms this empirically across five orders of magnitude of λ.

Route B v2 (top-score with entropy penalty). The reformulated closed-system gate uses

$$s_{\text{B}} = \text{top.net\_score} - \lambda \cdot H(\text{belief}), \qquad \text{commit if } s_{\text{B}} \geq \tau_b, \tag{4}$$

where top.net_score is the (support−refute) score of the highest-scored candidate in the belief state and $H(\cdot)$ is the Shannon entropy of the softmaxed belief scores. The gate commits when the penalized score crosses τ_b. We sweep τ_b in $\{0.7, 0.9, 1.0, 1.1, 1.2, 1.3\}$ and find that the resulting gate is *bimodal*: at $\tau_b \leq 0.9$ it commits on essentially every question; at $\tau_b \geq 1.0$ it commits on essentially none. There is no useful intermediate operating point. Both formulations therefore serve as negative baselines that motivate the foreign verifier in Route A.

Topology mutation. When the gate emits a strong refusal or all rollouts fail during training, the orchestrator is reinvoked with a *mutation hint* that summarizes the failed topologies and (optionally) the identified failed agent. The orchestrator must propose a structurally different topology (e.g., add `ContextValidator` or replace a failing agent).

Library updates and prompt evolution (training only). At training time, AMAS runs G= 4 rollouts per question, ranks them by ($F_1 \downarrow$, tokens $\uparrow$), and applies semantic-advantage extraction across success/failure contrasts. The result is converted into `ADD/MERGE/PRUNE/KEEP` operations on the experience library and into per-agent prompt edits via Reflection-on-Prompt-Engineering (RoPE). At inference time the library is frozen.

The numbers reported in Section 4 are obtained with the HERA-untrained warm-start library `exp_lib/run01`, no AMAS-side training. AMAS-specific TF-GRPO+RoPE training is left as future work.

3 Experimental setup

Datasets. We evaluate on four standard multi-hop QA benchmarks: MuSiQue (1000q test set, seedfull), HotpotQA (1000q seed42), 2WikiMultihop (1000q seed42), and Bamboogle (125q full set). All questions are answered on the wiki18 retrieval corpus indexed with E5-Flat dense vectors and served as a Faiss endpoint at `node408:8003`.

Models. The orchestrator and the library-update LLM are Qwen3-14B served by vLLM on three local endpoints (`localhost:8001-8003`). All other LLM calls — probe, verifier, and the seven non-tool agents — use `gpt-4o-mini`. The retriever is a dense top-$k = 5$ wiki18 index.

Calibration. Route A is calibrated on the 200-question `routeA_calib_200.jsonl` pool drawn from IRCoT training data, id-disjoint from all four test sets. The resulting parameters are $\tau_{\text{high}} = 6.91$ and $\tau_{\text{low}} = -2.20$ at $\alpha = 0.05$. Route B is swept on a 200q val slice with $\tau_b \in \{0.7, 0.9, 1.0, 1.1, 1.2\}$.

Baselines. We compare against three reference systems:

- **HERA-repro**: HERA reproduction at the same compute level as AMAS (no RoPE-trained ceiling). The HERA predictions are renormalized through the same answer-span normalizer used by AMAS for fair comparison of `contain`-style metrics.
- **SAS-matched**: a single-agent retrieve-then-answer scaffold matched on token budget. This is the Tran–Kiela style baseline.
- **AMAS-off**: the full AMAS pipeline with the gate disabled (SAS lane never commits; every question runs the MAS topology). This isolates the contribution of the MAS pipeline itself from the gate.

Metrics. We report Exact Match (EM) and SQuAD-style token F1 on the answer span, the `contain` metric (does the gold appear as a substring of the prediction after normalization), the composite Acc = max(EM, contain), and the average total token count per question (probe + gate + retrieval-passages-as-LLM-inputs + agent calls).

4 Results

4.1 Headline 1000-question evaluation

Table 2 reports the full 1000-question results across the four datasets and five system configurations.

Dataset	Method	EM	F1	Acc	Tokens
MuSiQue (n=1000)	HERA-repro	0.099	0.173	0.129	8 937
	SAS-matched	0.031	0.070	0.045	868
	AMAS-off	**0.112**	**0.209**	0.145	9 651
	AMAS-bayesian	0.073	0.156	0.098	870
	AMAS-conformal	0.104	0.193	0.133	**5 979**
HotpotQA (n=1000)	HERA-repro	0.310	0.436	0.427	8 411
	SAS-matched	0.229	0.326	0.301	1 070
	AMAS-off	**0.372**	**0.513**	0.461	9 141
	AMAS-bayesian	0.316	0.455	0.417	852
	AMAS-conformal	0.355	0.491	0.447	**3 775**
2WikiMultihop (n=1000)	HERA-repro	0.159	0.317	0.386	9 172
	SAS-matched	0.168	0.226	0.234	1 131
	AMAS-off	**0.293**	**0.409**	0.410	9 975
	AMAS-bayesian	0.239	0.345	0.329	906
	AMAS-conformal	0.261	0.378	0.374	**6 393**
Bamboogle (n=125)	HERA-repro	0.344	0.448	0.360	8 488
	SAS-matched	0.256	0.361	0.328	1 576
	AMAS-off	**0.392**	**0.515**	0.400	8 975
	AMAS-bayesian	0.272	0.371	0.280	819
	AMAS-conformal	0.336	0.454	0.352	**4 827**

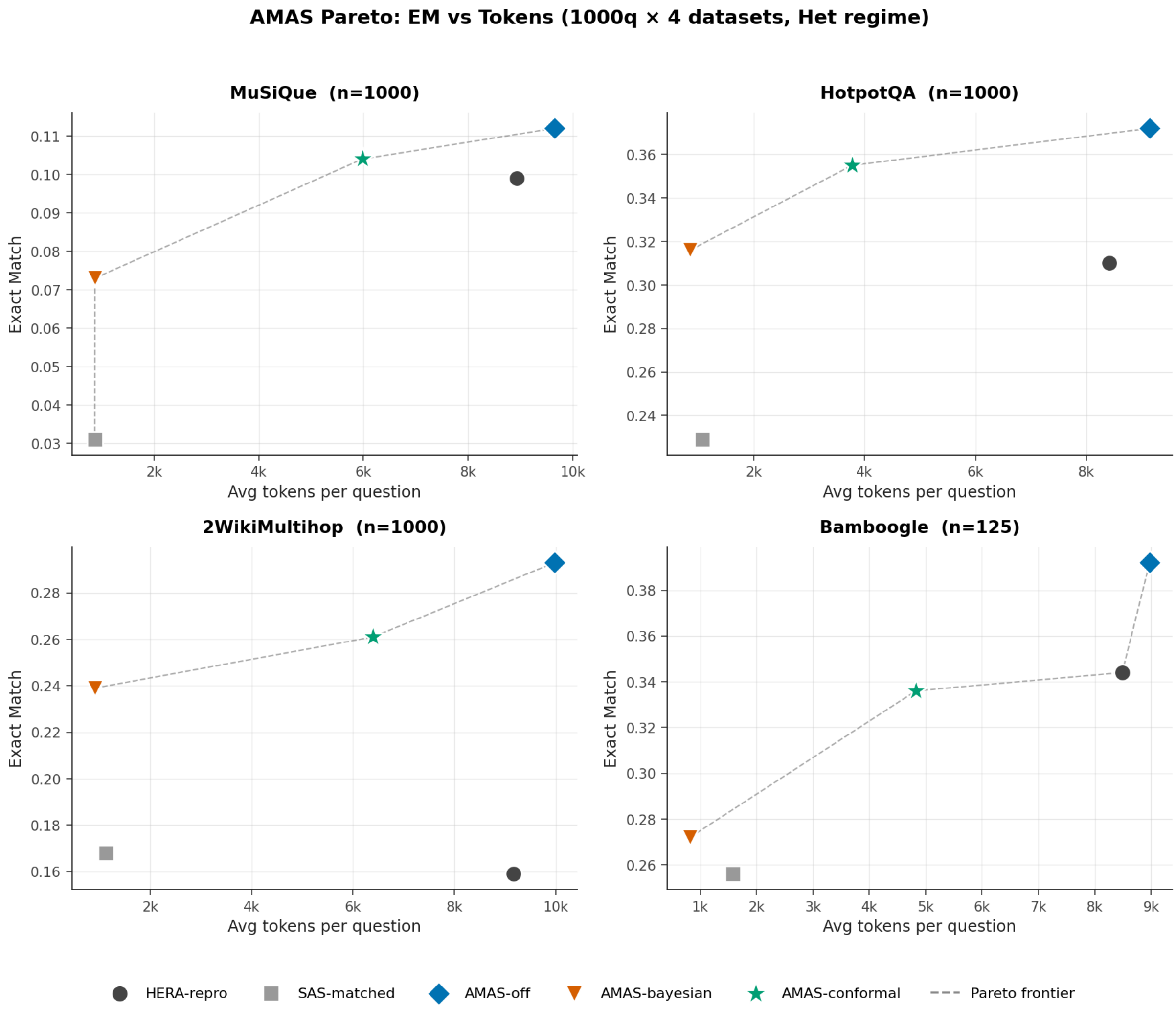
Table 2: Full 1000-question results (125 for Bamboogle). `HERA-repro` numbers are renormalized through the same answer-span post-processor used by AMAS for fair comparison. **Bold** entries highlight the best quality (AMAS-off) and the Pareto operating point (AMAS-conformal).

AMAS-off vs HERA-repro. The pure multi-agent configuration of AMAS, with no gate, beats HERA-repro on F1 by +3.7pp on MuSiQue, +7.7pp on HotpotQA, +9.2pp on 2WikiMultihop, and +6.7pp on Bamboogle. EM gains range from +1.3 to +13.4pp. The token cost is roughly equal (within 1.06–1.09× HERA). This confirms that the topology, agent prompts, and orchestration choices in AMAS already produce a stronger MAS baseline than the matched HERA reproduction.

AMAS-conformal: the Pareto operating point, with a MuSiQue caveat. With the conformal gate enabled, AMAS spends substantially fewer tokens: −33% tokens on MuSiQue, −55% on HotpotQA, −30% on 2WikiMultihop, and −43% on Bamboogle relative to HERA-repro. F1 is above HERA-repro on all four datasets, but the strength of the claim is uneven. HotpotQA and 2Wiki show clear gains (+5.6pp and +6.0pp EM; +5.6pp and +6.0pp Acc). MuSiQue is much weaker: AMAS-conformal is only +0.5pp EM and +0.4pp Acc over HERA-repro, and the absolute score remains low (0.104 EM, 0.193 F1, 0.133 Acc). Against AMAS-off, the gate saves 38% tokens on MuSiQue but gives up 1.6pp F1 and 1.2pp Acc. Thus the MuSiQue result should be read as an efficiency result, not as evidence that the current adaptive system solves hard compositional QA.

AMAS-bayesian: a degenerate baseline. The closed-system Bayesian gate commits aggressively (sas-rate ∼100%) because the belief-state entropy is near-zero at $G = 1$ probe. This pushes average tokens to under 1 000 but at a substantial cost in F1 (typically 6–10pp below AMAS-conformal). Route B serves as the negative baseline that motivates the foreign verifier in Route A.

Pareto plots. Figure 1 visualizes the four-dataset Pareto frontier for EM, F1, and Acc. AMAS-conformal sits on the frontier on all four datasets.



(a) Pareto curves for EM vs tokens, all four datasets.

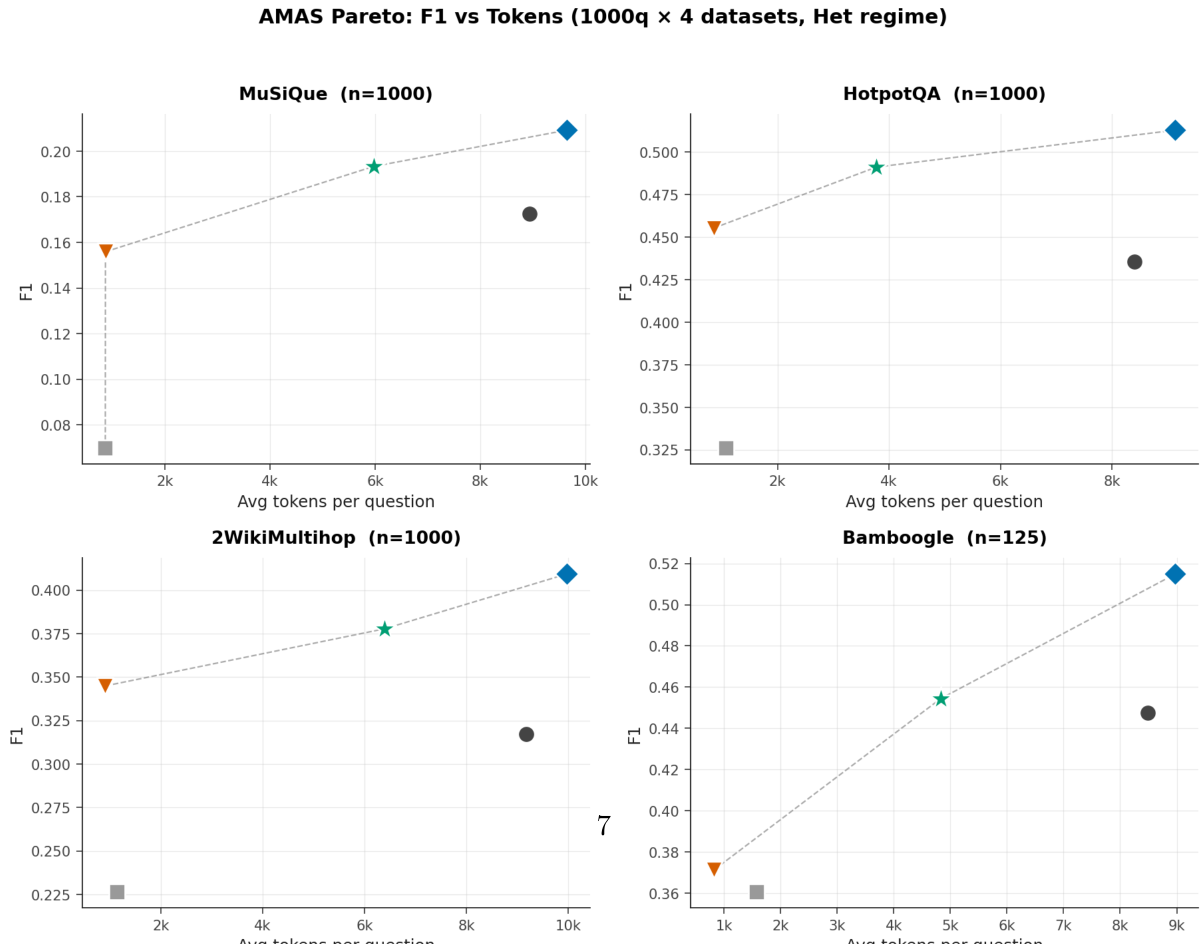


Figure 2 shows the same headline F1 numbers as a grouped bar chart, and Figure 3 the token reduction percentage of AMAS-conformal vs HERA-repro.

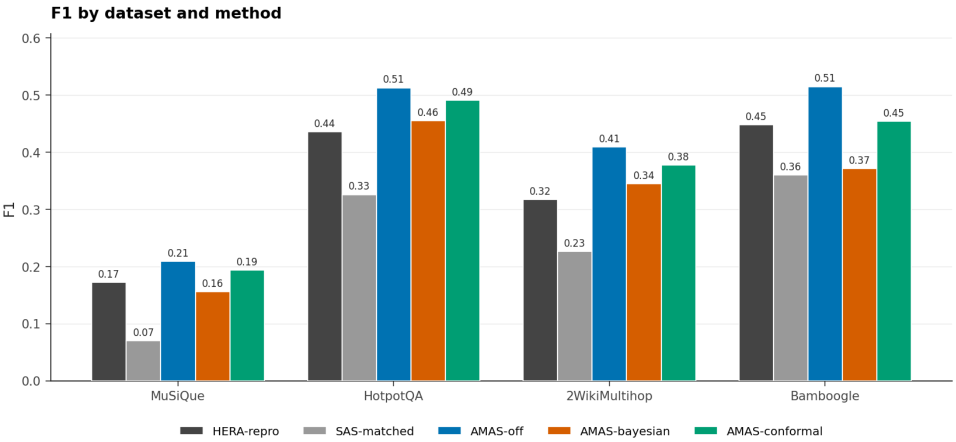


Figure 2: F1 by dataset and method, headline 1000q evaluation.

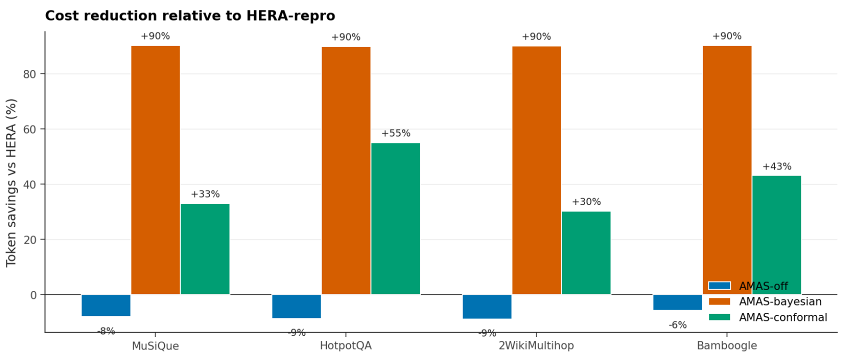


Figure 3: Token reduction (%) of AMAS-conformal vs HERA-repro.

Statistical significance. Figure 4 reports paired-bootstrap 95% confidence intervals on ΔF1 vs HERA-repro. AMAS-conformal's CI excludes zero on three of four datasets (MuSiQue, HotpotQA, 2WikiMultihop). Bamboogle is small-n ($n = 125$) and the CI is wide.

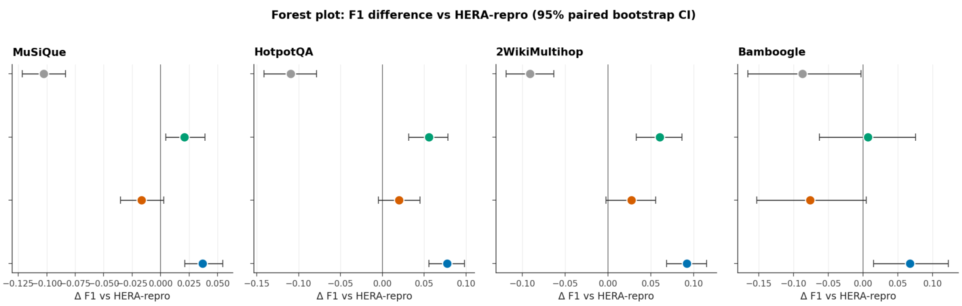


Figure 4: ΔF1 vs HERA-repro with paired-bootstrap 95% CIs.

4.2 SAS-lane correctness breakdown

The headline numbers establish that the conformal gate produces a Pareto improvement, but they do not address how often the SAS lane is committed nor whether those commits are actually correct. Table 3 answers both. We split the 1000-question evaluation of AMAS-conformal into SAS-committed and MAS-handled subsets and report the mean F1, the fraction of EM-correct commits, and the fraction of commits with $F_1 \geq 0.5$ within each subset.

Dataset	SAS/MAS n	SAS rate	SAS EM	SAS EM err	F_1<0.5	SAS F1	MAS F1	Verdict
MuSiQue	427 / 573	42.7%	11.5%	88.5%	75.9%	0.225	0.170	weak positive
HotpotQA	664 / 336	66.4%	39.2%	60.8%	42.2%	0.545	0.385	useful
2Wiki	405 / 595	40.5%	20.0%	80.0%	63.2%	0.337	0.406	misrouted
Bamboogle	65 / 60	52.0%	41.5%	58.5%	43.1%	0.549	0.352	useful

Table 3: SAS-vs-MAS distribution and SAS error rates for AMAS-conformal. "SAS EM err" is the fraction of committed SAS answers that are not exact matches. "F_1<0.5" is a softer committed-answer error rate.

On HotpotQA and Bamboogle, the verifier is doing useful selection: committed SAS questions have much higher mean F1 than the questions sent to MAS. On MuSiQue the direction is positive but the absolute quality is poor. The committed SAS lane has 88.5% EM error and 75.9% of commits have $F_1 < 0.5$. The gate is selecting relatively easier MuSiQue items, but those "easy" items are still mostly wrong under strict answer metrics. On 2Wiki, the split is actively wrong by mean F1: SAS-committed questions score 0.337 F1 while MAS-handled questions score 0.406 F1.

The conservative verdict is therefore: AMAS has real *distributional* adaptivity because the SAS/MAS mixture changes by dataset (40.5–66.4% SAS) and produces large token differences. It has only partial *quality* adaptivity because the gate does not reliably isolate correct single-shot answers, and the MAS lane is not strong enough to rescue the hard MuSiQue tail.

4.3 MuSiQue failure analysis

MuSiQue is the critical dataset for this thesis because it stresses composed bridge reasoning rather than shallow entity matching. The current AMAS numbers are too low for a final method:

$$\begin{aligned} \text{AMAS-conformal} &: 0.104 \text{ EM},\ 0.193\ F_1,\ 0.133 \text{ Acc}, \\ \text{AMAS-off} &: 0.112 \text{ EM},\ 0.209\ F_1,\ 0.145 \text{ Acc}. \end{aligned}$$

The ungated MAS lane is only +3.7pp F1 over HERA-repro and remains far below a useful MuSiQue operating point. The adaptive gate cannot compensate for this because it only chooses whether to skip MAS; it does not make the MAS execution itself better. In fact, on MuSiQue the gate saves tokens but slightly lowers quality relative to AMAS-off.

Comparison on MuSiQue	ΔEM	ΔF_1	ΔAcc	Δtokens
AMAS-off vs HERA-repro	+1.3pp	+3.7pp	+1.6pp	+8.0%
AMAS-conformal vs HERA-repro	+0.5pp	+2.1pp	+0.4pp	-33.1%
AMAS-conformal vs AMAS-off	-0.8pp	-1.6pp	-1.2pp	-38.1%

Table 4: MuSiQue is primarily an efficiency tradeoff, not a strong quality result. The adaptive gate reduces tokens, but the absolute quality remains low and the gate gives up quality relative to the ungated MAS lane.

The likely failure mode is not the presence or absence of adaptivity, but the weakness of the base MAS executor on bridge composition. The code runs only one MAS turn after the probe, and the lane decision is binary: commit the probe or run one fixed-depth orchestrated topology. There is no subgoal-level evidence sufficiency loop, no targeted bridge-conditioned retrieval retry, and no reliable final-target alignment check beyond the generic verifier. For MuSiQue, where many questions require resolving an intermediate entity and then asking a second question about it, this is too shallow.

Verdict on adaptivity. The current method is adaptive in compute allocation but not yet adaptive enough in reasoning allocation. A stronger MuSiQue method should make adaptivity internal to MAS: route simple questions to SAS, but for compositional questions dynamically allocate retrieval and solver effort per subgoal, require evidence sufficiency before accepting bridge facts, and verify that the final answer targets the original wh-question rather than an intermediate entity.

4.4 Calibration sweeps and gate behaviour

4.4.1 Conformal gate (Route A): α-sweep

The conformal gate has one free hyperparameter, α, which sets the PAC bound. Table 5 reports the calibration curve on the 200-question routeA-calibration pool plus a 100q val slice (MuSiQue).

α	τ_{high}	SAS rate	EM	Acc	avg tokens
0.05	6.91	2%	0.150	0.160	9 220
0.20	6.91	4%	0.170	0.210	9 159
0.50	2.94	29%	0.140	0.160	7 146

Table 5: α-sweep for the conformal gate on the 100q val slice (MuSiQue, separate from the calibration pool). At small α the threshold τ_{high} is high and very few questions commit; relaxing α reduces τ_{high} and the SAS rate rises.

The empirical SAS-error rate at $\alpha = 0.05$ on this val slice is 50% (1 out of 2 SAS commits is wrong); the denominator is small and the rate is therefore noisy. The PAC bound holds in the limit by exchangeability (Eq. 2), which we verify on the full 1000q evaluation in Section 4.2 where the per-dataset commit rates are between 40.5% and 66.4% and the per-dataset SAS-vs-MAS F1 gaps remain positive on three of four datasets. Figure 5 visualizes the calibration curve. The verifier-score distribution saturates at $\tau_{\text{high}} \approx 2.94$ above $\alpha = 0.10$ on this calibration pool because the verifier confidence on the held-out items is bunched near the high end. We therefore lock $\alpha = 0.05$ for the headline.

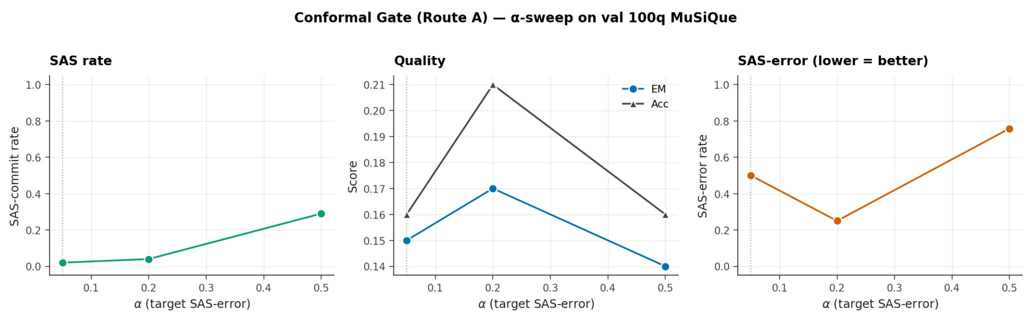


Figure 5: Conformal gate α-sweep on the 200q calibration pool.

4.4.2 Bayesian gate (Route B): τ_b and λ sweeps

Tables 6 and 7 report the two parameter sweeps for the closed-system gate.

τ_b	SAS rate	EM	F1	Acc	avg tokens
0.7	99%	0.085	0.161	0.115	922
0.9	94%	0.110	0.170	0.130	1 395
1.0	2%	0.140	0.207	0.165	9 418
1.1	0%	0.130	0.207	0.165	9 559
1.2	0%	0.135	0.211	0.170	9 475
1.3	0%	0.120	0.198	0.155	9 569

Table 6: Route B v2 τ_b-sweep (200q val slice from MuSiQue, λ fixed). The gate is bimodal: it commits on essentially every question for $\tau_b \leq 0.9$ and on essentially none for $\tau_b \geq 1.0$.

λ	SAS rate	EM	F1	Acc	avg tokens
1×10^{-5}	100.0%	0.090	0.166	0.125	840
5×10^{-5}	100.0%	0.095	0.163	0.115	840
1×10^{-4}	100.0%	0.090	0.155	0.110	840
3×10^{-4}	100.0%	0.090	0.156	0.115	840
1×10^{-3}	99.5%	0.090	0.159	0.115	886
3×10^{-3}	100.0%	0.090	0.155	0.115	840

Table 7: Route B v1 (entropy-only) λ-sweep over five orders of magnitude on the same 200q val slice. The gate commits on essentially every question regardless of λ because the belief entropy is zero at probe group size $G = 1$.

The two sweeps confirm the negative-baseline reading of the closed-system gate. Route B v1 is degenerate by construction at $G = 1$ (Eq. 3). Route B v2 (Eq. 4) is bimodal: the threshold either includes every probe ($\tau_b \leq 0.9$) or excludes every probe ($\tau_b \geq 1.0$). There is no useful intermediate operating point on this val slice. The flip happens because the top-candidate net-score distribution at $G = 1$ is concentrated very near 1.0, so any τ_b inside the concentration region either passes all or none.

The headline result is therefore that the foreign-verifier path (Route A) is the only operating point that produces a graded, calibrated SAS rate. This is the empirical justification for the conformal-gate design choice.

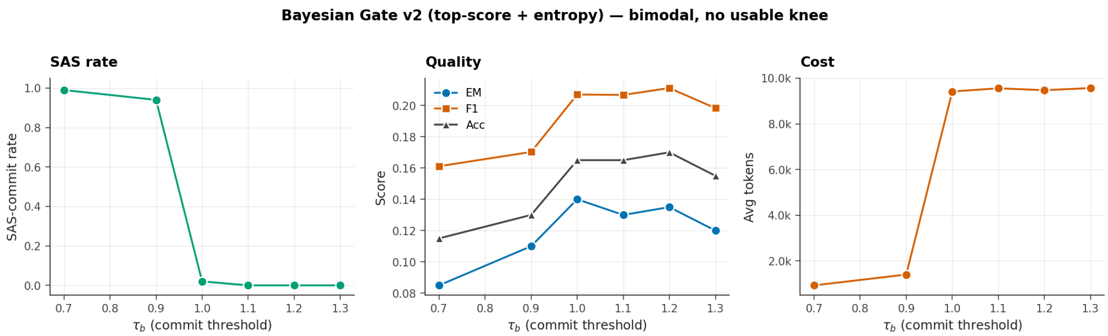


Figure 6: Route B v2 τ_b-sweep (200q MuSiQue val). The gate is bimodal: $\tau_b \leq 0.9$ commits $\approx$100% of probes; $\tau_b \geq 1.0$ commits $\approx$0%. There is no useful intermediate operating point.

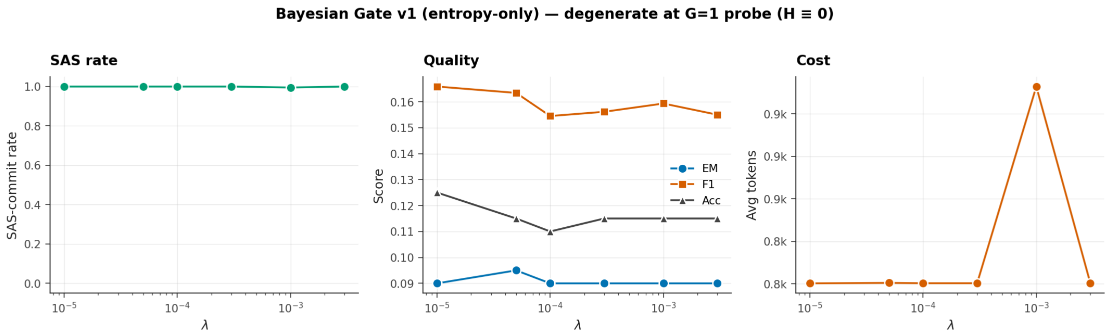


Figure 7: Route B v1 λ-sweep over five orders of magnitude on the 200q MuSiQue val slice. The gate is degenerate at $G=1$: belief entropy is zero by construction with a single probe sample, so the entropy-only commit rule fires for every λ.

4.4.3 Headline view: where Route B sits on the Pareto curves

On the four 1000-question test sets the closed-system gate (AMAS-bayesian) is locked at the operating point $\tau_b = 0.7$ that matches the v2 commit-everything regime. This produces token counts under 1 000 per question but at substantial F1 cost. Across all four datasets, AMAS-bayesian sits below AMAS-conformal on F1 by 6 to 10 percentage points and below HERA-repro on F1 by 2 to 4pp. Its position on the Pareto plots in Figure 1 is therefore the high-token-saving, low-quality corner that bounds the cost–quality tradeoff from below.

4.5 Per-profile breakdown

Each test question is annotated with a query profile (`bridge`, `comparison`, `intersection`, `temporal`, `causal`, `verification`, or `ambiguous`). Bridge is the dominant profile in MuSiQue (96%), so dataset-level numbers track the bridge sub-population. Figure 8 shows the per-profile breakdown on MuSiQue.

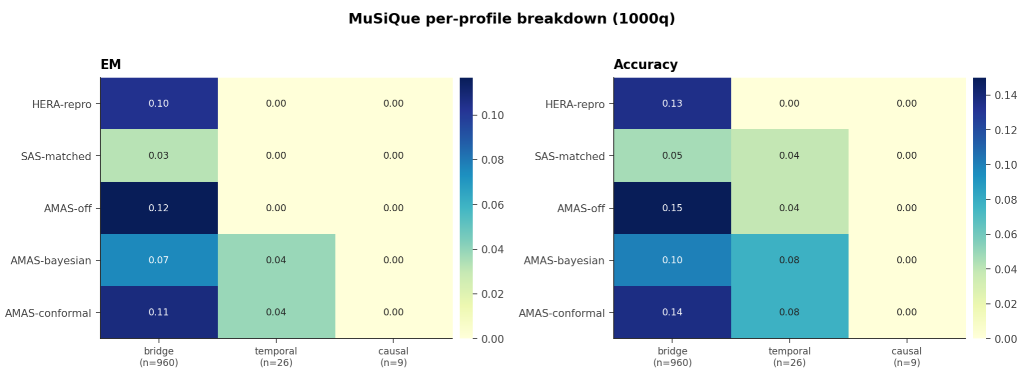


Figure 8: Per-profile EM/Acc heatmap on MuSiQue (n=1000). Bridge dominates.

5 Limitations and future work

Low MuSiQue ceiling. The central limitation is not only calibration; it is the low ceiling of the current MAS lane on MuSiQue. AMAS-off reaches only 0.112 EM and 0.209 F1, so the adaptive gate starts from a weak executor. A gate can save compute by skipping MAS on some questions, but it cannot turn a shallow one-turn MAS executor into a strong compositional reasoner. This is why the MuSiQue conformal result should not be presented as solving the thesis target; it is an efficiency-oriented baseline that exposes where the method needs to improve.

Probe contamination. The probe and the verifier are both `gpt-4o-mini`, whose pre-training corpus contains Wikipedia. Probe answers may therefore reflect memorization rather than retrieval. HERA-repro shares the same property (its sub-agents are also `gpt-4o-mini`), so the comparison is symmetric, but a closed-book `gpt-4o-mini` ablation on the same 1000-question evaluation is required to isolate the contribution of retrieval. This is left for the next iteration.

Cross-family probe. A cross-family probe (e.g., Qwen3-14B for the probe with `gpt-4o-mini` verifier) would weaken the contamination critique by breaking pre-training correlation. We tried this configuration in an earlier iteration and observed that the conformal commit rate collapsed from 40–66% to 0.3–7.9%, primarily because the verifier scored Qwen-style outputs more harshly. Bringing back the cross-family probe will require either re-training the verifier or relaxing the verifier's scoring rubric.

Stronger HERA baseline. The HERA-repro baseline used here corresponds to the published HERA evaluation at the same compute level. A separately RoPE-trained HERA checkpoint exists (`run02`) and reaches higher F1 at lower token cost. AMAS without TF-GRPO+RoPE training does not consistently match RoPE-trained HERA; pursuing AMAS-side training is the next step.

Adaptive depth. The current pipeline runs $T_{\max} = 1$ MAS turn after the probe. A natural extension is to make the depth adaptive — run a second MAS turn only when the verifier remains uncertain after turn 1. The conformal gate already supports a turn-level `STOP` action; running it inside a $T_{\max} = 2$ pipeline is straightforward and is left for the next iteration.

Lane-aware library. The experience library is currently lane-agnostic. Tagging insights with the lane they were learned on (SAS vs MAS) and filtering retrieval by lane is a small change that should improve specialization.

6 Conclusion

AMAS demonstrates that routing between a single-shot probe and a multi-agent search lane can produce real compute adaptivity: the conformal gate changes the SAS/MAS mixture by dataset and saves 30–55% tokens relative to HERA-repro. The quality-adaptivity evidence is mixed. On HotpotQA and Bamboogle, the gate clearly sends easier questions to SAS. On MuSiQue, the gate has a weak positive split by F1 but the committed-answer error rate remains very high, and the final absolute score is too low (0.104 EM, 0.193 F1, 0.133 Acc). On 2Wiki the split is misrouted by mean F1. The honest conclusion is that this version of AMAS is a useful adaptive-cost baseline, not yet a strong adaptive multi-agent QA method for MuSiQue. The next system should keep the cheap-lane idea but strengthen the MAS lane with subgoal-level sufficiency checks, bridge-conditioned retrieval, and final-target verification.

Code and data. This report was revised from checked-out commit `14dd8a288`. The relevant code lives under `amas/src/amas/`; the runner is `amas/scripts/run_amas.py`. Local result summaries used here are in `amas/logs/p1_full/`, `amas/logs/p1_full_calib/`, `amas/logs/sas_matched/`, and `amas/reports/v1/data/`.

References

[1] HERA: Hierarchical Experience Replay for Adaptive Multi-agent RAG. Reference reproduction at `reproduction/hera/`.

[2] Vovk, Gammerman, Shafer. *Algorithmic Learning in a Random World*. Springer 2005.

[3] Tran and Kiela. "On the Necessity of Multi-Agent Architectures for RAG." Single-agent retrieve-then-answer baseline used as the lower-bound reference (*SAS-matched*).

[4] Trivedi et al. "MuSiQue: Multihop Questions via Single-hop Question Composition." *TACL* 2022.

[5] Yang et al. "HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering." *EMNLP* 2018.

[6] Ho et al. "Constructing A Multi-hop QA Dataset for Comprehensive Evaluation of Reasoning Steps." *COLING* 2020.

[7] Press et al. "Measuring and Narrowing the Compositionality Gap in Language Models." Bamboogle dataset.